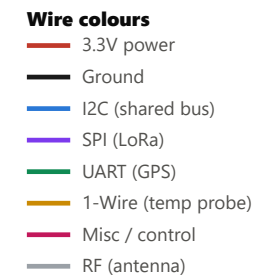


Real components, real pin-level wiring — matches hardware/HARDWARE_DESIGN.md \$1.3 exactly. Build on perfboard or dupont-wire breadboard before committing to a PCB order.



Build notes: Power rails (red 3V3 / black GND) are drawn as horizontal buses — every component's power pin taps into the nearest point on the rail, exactly as you'd wire a breadboard's rail strips. I2C (blue) is a shared bus: MPU-6050 and MAX30102 both tap the same SD A/SCL lines feeding the ESP32, with 4.7kΩ pull-ups to 3V3. SPI (purple) is dedicated point-to-point wiring to the LoRa module only. Full BOM with part numbers and supplier links: [BUDGET_PROPOSAL.md](#) and [hardware/actual_equipment/README.md](#).